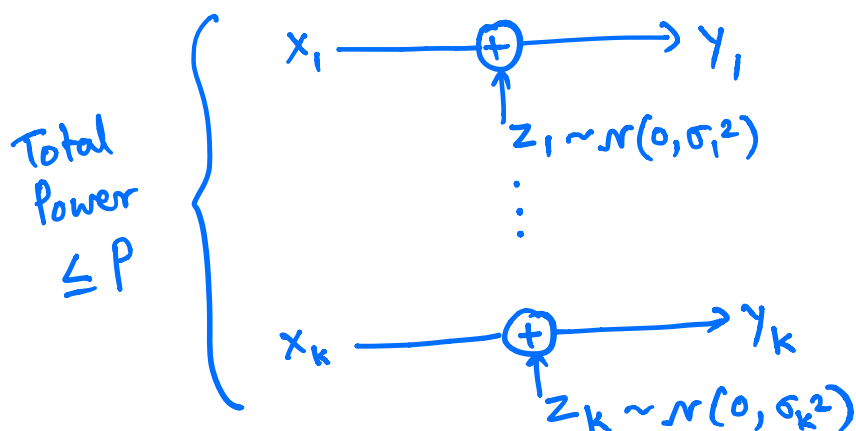


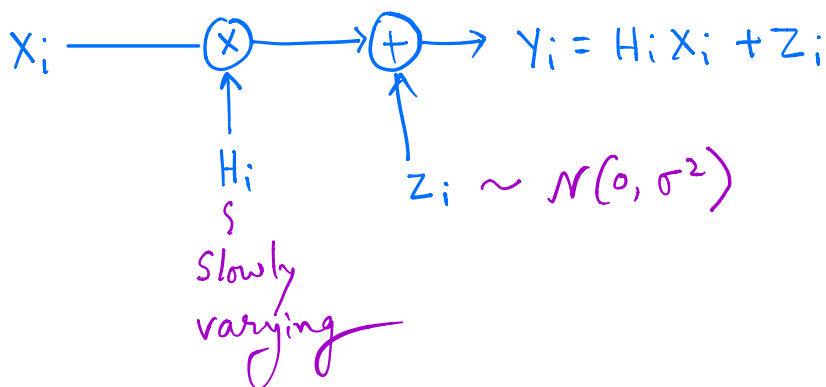
Parallel Gaussian channels:



$$y_j = x_j + z_j \quad ; \quad j=1, 2, \dots, k$$

$$\sum_{j=1}^k \sum_{i=1}^n x_{ji}^2 \leq nP$$

Applications: ① Fading channel

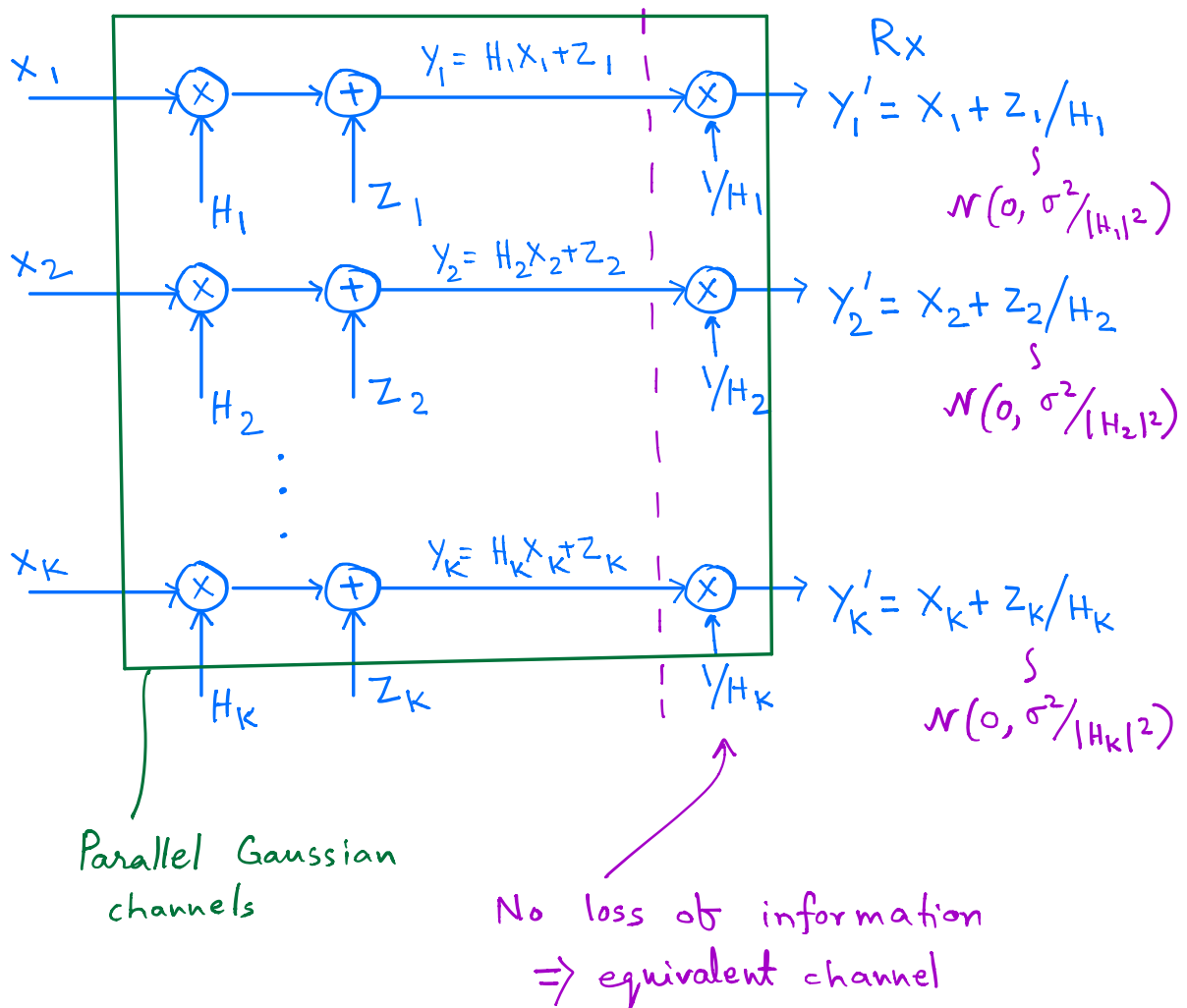


Block fading :

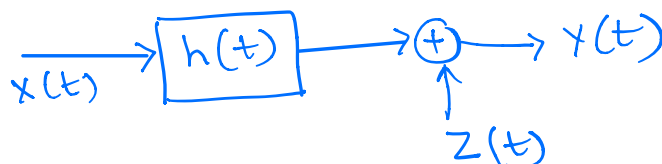
H_i is fixed in a block of length n

- changes i.i.d from block to block.

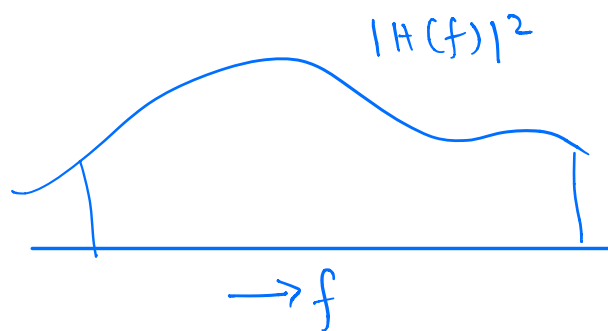
- known at the Rx and Tx.



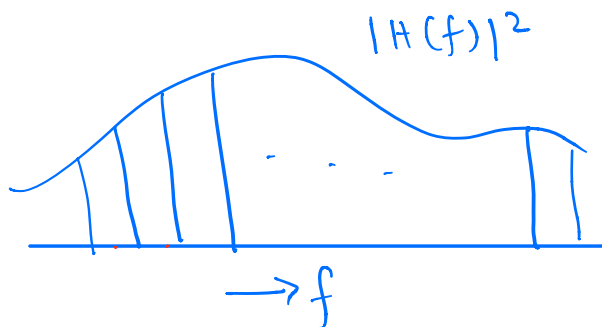
② OFDM



channel freq.
response



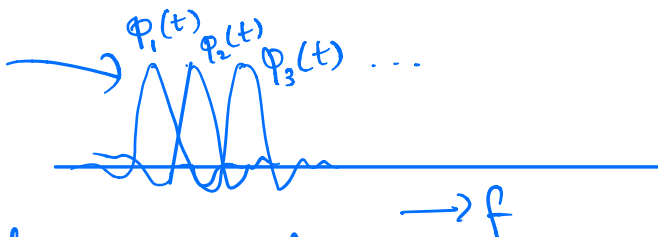
- results in signal distortion
- equalization results in colored noise
- Single carrier modulation distributes information in all freq. irrespective of $H(f)$.



orthogonal
carriers



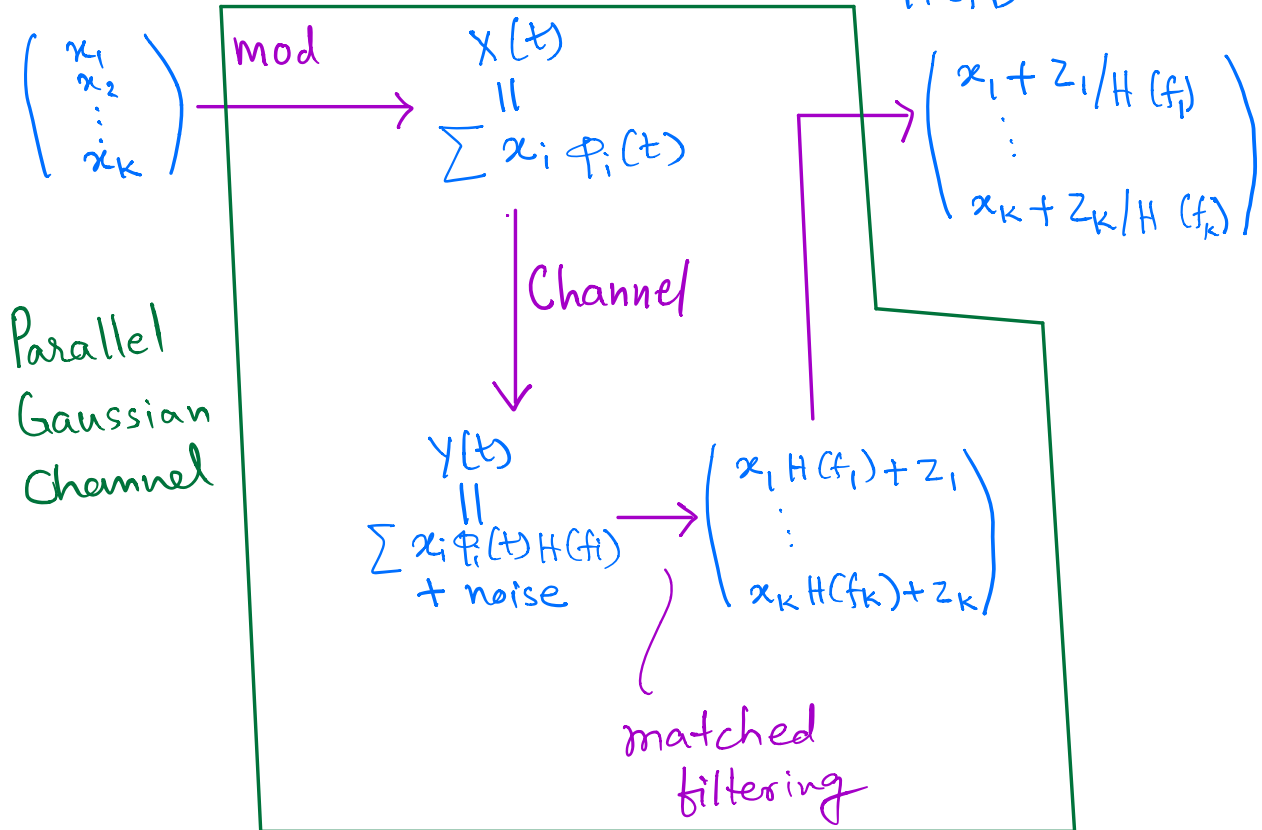
go through orthogonal
channels → independent noise



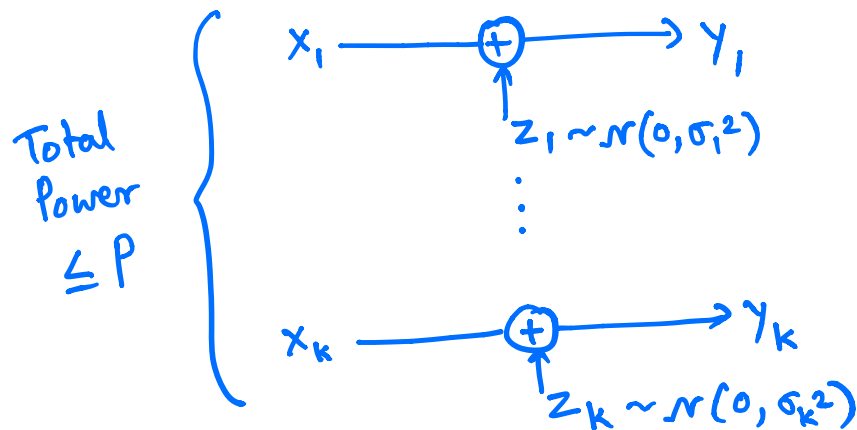
2

different attenuation

$H(f_i)$



Parallel Gaussian channels:



$$y_j = x_j + z_j \quad ; \quad j=1, 2, \dots, k$$

$$\sum_{j=1}^k \sum_{i=1}^n x_{ji}^2 \leq nP$$

Information capacity:

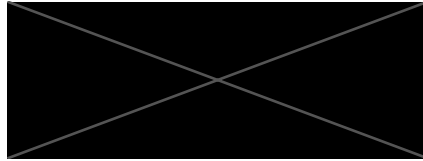
$$C = \max_{f(x_1, \dots, x_k): \sum_j E x_j^2 \leq P} I(x_1, \dots, x_k; y_1, \dots, y_k)$$

$$\leq \max_{\sum_j p_j \leq P} \sum_j \frac{1}{2} \log \left(1 + \frac{p_j}{\sigma_j^2} \right) \quad (\text{See Cover \& Thomas for proof})$$

Using similar steps as in the converse for AWGN channels,

$$nR \leq n \sum_j \frac{1}{2} \log \left(1 + \frac{P_j}{g_j} \right) \quad \text{--- ①}$$

where $P_j = E X_j^2$



By the power constraint,

$$\sum_j P_j \leq P.$$

"=" is achieved when x_j 's are i.i.d. $\mathcal{N}(0, P_j)$

So C is achieved by maximizing ①
subject to $\sum P_j = P$, $P_j \geq 0 \forall j$.

Using Lagrange multiplier method:

(1)

$$\underline{P} = (P_1, \dots, P_k)$$

$$\text{To minimize } f(\underline{P}) = -\sum_{j=1}^k \frac{1}{2} \log \left(1 + \frac{P_j}{\sigma_j^2} \right)$$

$$\text{under } \sum P_j \leq P$$

$$P_j \geq 0 \text{ for } j=1, 2, \dots, k$$

Clearly the optimum value $f^*(\underline{P})$ is a strictly ~~monotone~~ decreasing w.r.t. P , since the feasible region is strictly increasing as a fnc. of P and $f(\underline{P})$ is componentwise strictly decreasing w.r.t. $\underline{P}$. Hence the soln is the same if we replace $\sum P_j \leq P$ with equality condition $\sum P_j = P$. Hence we have the new optimization problem:

$$\min f(\underline{P}) = -\sum \frac{1}{2} \log \left(1 + \frac{P_j}{\sigma_j^2} \right)$$

$$\text{under } \sum P_j = P$$

$$-P_j \leq 0 \text{ for } j=1, 2, \dots, k$$

We write the Lagrangian:

$$J = -\sum \frac{1}{2} \log \left(1 + \frac{P_j}{\sigma_j^2} \right) + \lambda (\sum P_j - P) + \sum \mu_j P_j$$

(2)

Note that the feasible region has a non-empty interior. Hence Slater's condition is satisfied.

Hence strong duality holds and thus

$\underline{P}^*$ is an optimal solution if and only if $\exists v^*$ and $(\mu_1^*, \dots, \mu_k^*)$ which satisfy the KKT conditions:

$$\frac{\partial J}{\partial P_j} = 0 \quad \text{for } j=1, 2, \dots, k \quad \text{--- (1)}$$

$$\sum P_j = P \quad \text{--- (2)}$$

$$\sum P_j = P \quad \text{--- (3)}$$

$$-P_j \leq 0$$

$$\mu_j \geq 0 \quad \text{for } j=1, 2, \dots, k \quad \text{--- (4)}$$

and $\mu_j P_j = 0$ " (complementary slackness condition)
 --- (5)

Solution: (1) $\Rightarrow -\frac{1}{2} \cdot \frac{1}{\sigma_j^2 + P_j} + v - \mu_j = 0$ for $j=1, 2, \dots, k$

$$\Rightarrow 2(\sigma_j^2 + P_j) = -\frac{1}{v - \mu_j}$$

$$\Rightarrow P_j = -\frac{1}{2v} - \sigma_j^2$$

$$\text{defining } \lambda = -\frac{1}{2v}$$

$$-\frac{1}{2} \cdot \frac{1}{\sigma_j^2 + P_j} + v - \mu_j = 0$$

$$\Rightarrow \frac{1}{2(\sigma_j^2 + P_j)} = v - \mu_j$$

$$\Rightarrow P_j = \frac{1}{2(v - \mu_j)} - \sigma_j^2 \quad \text{--- (6)}$$

Case I: $\mu_j = 0$

$$P_j = \frac{1}{2\nu} - \sigma_j^2$$

$$= \lambda - \sigma_j^2 \quad \text{where } \lambda := \frac{1}{2\nu}$$

Since $P_j \geq 0$, ~~$\lambda \geq \sigma_j^2$~~ $\lambda \geq \sigma_j^2 > 0$

Case II: $\mu_j > 0$

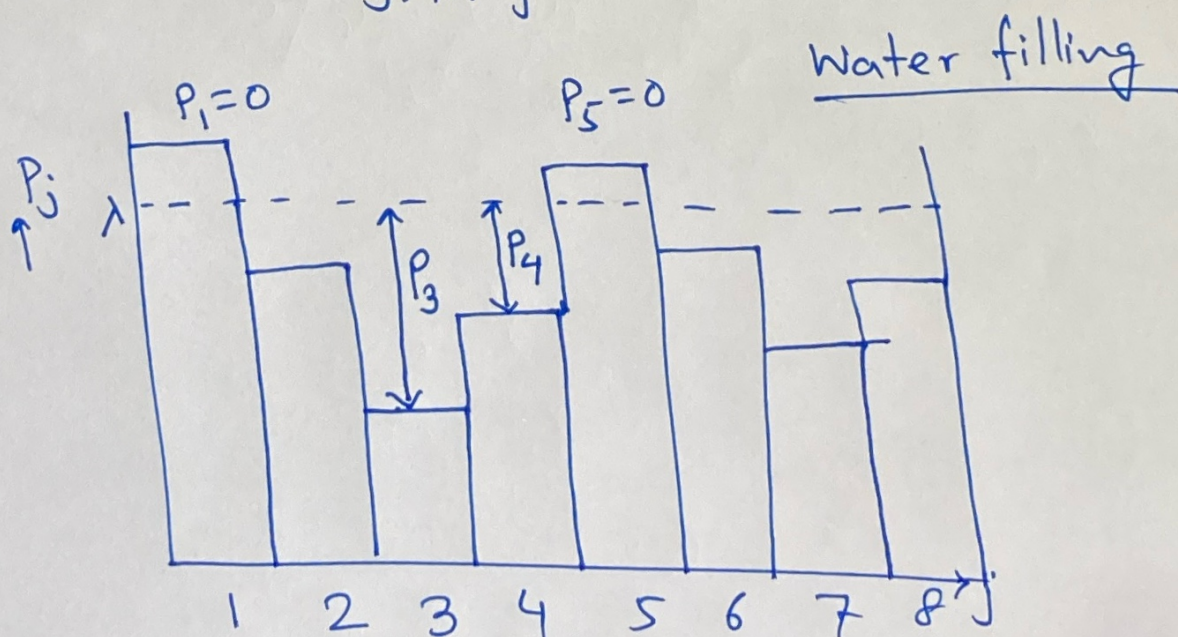
⑤ implies $P_j = 0$

Then ⑥ implies $\sigma_j = \frac{1}{2(\nu - \mu_j)} > \frac{1}{2\nu}$ (since $\mu_j > 0$)
 $= \lambda$

So
$$P_j = \begin{cases} \lambda - \sigma_j^2 & \text{if } \sigma_j^2 \leq \lambda \\ 0 & \text{otherwise} \end{cases}$$

$$= (\lambda - \sigma_j^2)^+$$

λ is s.t. $\sum_{j: \lambda > \sigma_j} (\lambda - \sigma_j^2)^+ = P$



if $S \subseteq \{1, \dots, k\}$ is the active channels,

then

$$p_j = \begin{cases} \nu - \sigma_j^2 & \text{for } j \in S \\ 0 & \text{for } j \notin S \end{cases}$$

$$P = \sum_{j \in S} p_j = |S|\nu - \sum_{j \in S} \sigma_j^2$$

$$\Rightarrow \nu = \frac{P + \sum_S \sigma_j^2}{|S|}$$

Algorithm:

$$S = \{1, 2, \dots, k\}$$

$$\nu = \frac{P + \sum_S \sigma_j^2}{k}$$

While $\nu < \sigma_j^2$ for any $j \in S$

For each $j \in S$:

remove j from S

if $\nu < \sigma_j^2$

$$\nu = \frac{P + \sum_S \sigma_j^2}{|S|}$$

We will now prove that the channels removed in one iteration will not "deserve" to become active in subsequent iterations.

$$\nu = \frac{P + \sum_{i=1}^k \sigma_i^2}{k}$$

Suppose, the surviving channels in the beginning of current iteration are $1, \dots, k$
 Wlog, suppose $\sigma_1^2 \leq \sigma_2^2 \leq \dots \leq \sigma_k^2$

Suppose j is s.t. $\sigma_j^2 \leq \nu < \sigma_{j+1}^2$

Then the channels $j+1, j+2, \dots, k$ are removed in the current iteration.

We denote $S_k := \sum_{i=1}^k \sigma_i^2$

In the next iteration, $\nu' = \frac{P + S_j}{j}$

$$\text{Now, } \nu = \frac{P + S_k}{k} = \frac{P + S_j + \sum_{i=j+1}^k \sigma_i^2}{k}$$

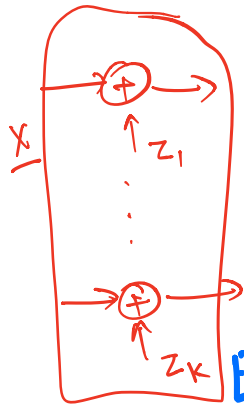
$$\geq \frac{P + S_j}{k} + \frac{(k-j)\nu}{k} \quad \text{since } \nu < \sigma_i^2 \quad \forall i > j$$

$$\Rightarrow \frac{j}{k} \nu \geq \frac{P + S_j}{k}$$

$$\Rightarrow \nu \geq \frac{P + S_j}{j} = \nu' \Rightarrow \nu' < \sigma_i^2 \quad \forall i > j$$

Hence the value of ν decreases with iterations, and hence the ^{noise variance in the} removed channels ~~remain~~ ^{remain} higher than the ~~next~~ ^{subsequent} levels ν' . So these channels will deserve to be inactive ~~even~~ in subsequent steps.

Parallel channels w. colored Gaussian noise:



$$\underline{x} \rightarrow \oplus \rightarrow \underline{y} = \underline{x} + (\underline{z} - \underline{\mu}_z)$$

$$\underline{z} \sim \mathcal{N}(\underline{0}, \underline{K}_z)$$

$$E[\underline{z}\underline{z}^T] = \underline{K}_z = \underline{Q}\underline{Q}^T$$

$$\text{where } \underline{Q}\underline{Q}^T = \underline{I}$$

$$\left[\begin{array}{l} \text{Input power constraint:} \\ \sum_i \sum_j x_{ji}^2 \leq nP \end{array} \right]$$

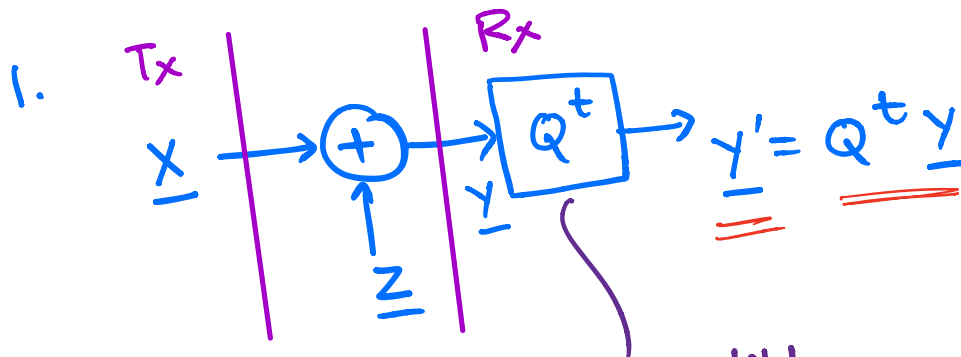
$$C = \max_{f(\underline{x}):} \underline{I}(\underline{x}; \underline{y})$$

$$\frac{1}{n} \sum_i E x_i^2 \leq P$$

|||

$$\frac{1}{n} \text{tr}(\underline{K}_x) \leq P$$

Transformations on the channel:



invertible
 $\Rightarrow$ does not change capacity, and
 $\underline{y}'$ is a suff. stat. for $\underline{y}$

$$\underline{y}' = Q^t \underline{x} + \boxed{Q^t \underline{z}}$$

$\underline{z}'$

$$K_{\underline{z}'} = E[\underline{z}' \underline{z}'^t]$$

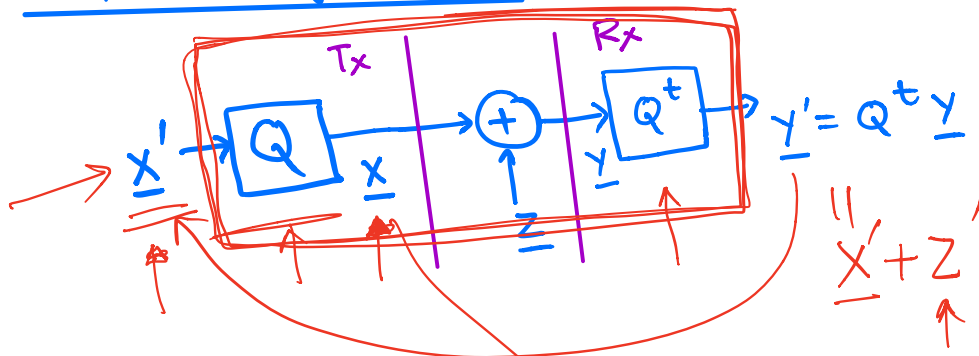
$$= Q^t E[\underline{z} \underline{z}^t] Q$$

$$= \Lambda$$

$$K_{\underline{z}} = Q \Lambda Q^t$$

$\rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

2. Preprocessing at Tx



$$\begin{aligned}\underline{y}' &= \underline{Q}^t \underline{x} + \underline{z}' \\ &= \underline{Q}^t \underline{Q} \underline{x}' + \underline{z}' \\ &= \underline{x}' + \underline{z}'\end{aligned}$$

$$\underline{K}_x = \begin{bmatrix} \cdot & & \\ & \ddots & \\ & & \cdot \end{bmatrix} \leq P$$

$$\underline{\text{Tr}}(\underline{K}_x) \leq P$$

$$\begin{aligned}\underline{K}_{x'} &= E[\underline{x}' \underline{x}'^t] \\ &= \underline{Q}^t \underline{K}_x \underline{Q}\end{aligned}$$

$$\begin{aligned}\text{Tr}(\underline{Q}^t \underline{K}_x \underline{Q}) &= \text{Tr}(\underline{K}_x \underline{Q} \underline{Q}^t) \\ &= \text{Tr}(\underline{K}_x) \mathbb{I}\end{aligned}$$

$$\underline{\text{Tr}}(\underline{K}_{x'}) = \underline{\text{Tr}}(\underline{K}_x)$$

Using relation:
 $\underline{\text{tr}}(\underline{AB}) = \underline{\text{tr}}(\underline{BA})$

So, power constraint P on $\underline{x}$
 $\Leftrightarrow$ " " " P " $\underline{x}'$

Suppose:

$$\underline{K}_{z'} = \underline{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_k)$$

Then, power allocation can be done
by water filling as

$$\underline{x}' \rightarrow \underline{y}'$$

$$P_j = (v - \lambda_j)^+$$

where v is s.t.

$$\sum (v - \lambda_j)^+ = P$$